

SAMS Operations Runbook

Smart Attendance Management System — Super Admin troubleshooting blueprint

Item	Value
Production app	https://app.smart-managment.com
Super Admin	https://super.smart-managment.com (school code SUPERADMIN)
API (direct)	https://api.smart-managment.com
VPS path	/var/www/sams
API port (local)	3001
PM2 app name	sams-api

Use this when something breaks: **symptom** → **where to check** → **commands** → **fix**.

VPS one-shot scripts

For repeatable deploy and smoke tests after `git pull`, use the scripts in [VPS-SCRIPTS.md](#):

```
cd /var/www/sams && git pull origin main && bash scripts/vps-import-bundle.sh
```

1. First response (any outage)

```
cd /var/www/sams
pm2 status
curl -sS http://127.0.0.1:3001/health
sudo nginx -t
```

Health field	Meaning
checks.database: true	PostgreSQL OK
checks.redis: true	Redis OK
ai.configured: true	AI keys loaded
sms.configured: true	Africa's Talking keys loaded

If `/health` fails locally → **backend** issue (PM2, env, DB, Redis).

If local OK but browser fails → **nginx** or **Cloudflare/DNS**.

2. Site down / blank page / 502

Symptom: White screen or 502 Bad Gateway

Check: PM2 running?

```
pm2 status
pm2 logs sams-api --lines 50 --nostream
```

Fix — start API:

```
cd /var/www/sams
pm2 start ecosystem.config.js --env production --update-env
pm2 save
bash scripts/restart-api.sh
```

Check: nginx:

```
sudo nginx -t
sudo systemctl status nginx
sudo systemctl reload nginx
```

Check: Frontend built?

```
ls packages/frontend/dist/index.html
ls packages/super-admin/dist/index.html
```

Fix — rebuild UI:

```
cd /var/www/sams
npm run build -w @sams/frontend
npm run build -w @sams/super-admin
```

3. PM2 empty after deploy

Symptom: `pm2 status` shows no `sams-api` after `deploy-production.sh`

Cause: Often junk line in `secrets/providers.env` (e.g. pasted `bash scripts/restart-api.sh#`) executed during env load.

Check:

```
grep -n "restart-api\\|bash \\|pm2 \\|curl " secrets/providers.env packages/backend/.env
```

Fix env file: Only `#` comments and `KEY="value"` lines. Then:

```
bash scripts/restart-api.sh
```

Never paste: `bash scripts/restart-api.sh#` (trailing `#` breaks the path).

4. Startup secret / API crash loop

Symptom: Logs show `JWT_SECRET` , `JWT_REFRESH_SECRET` , `QR_SECRET` , or `LICENSE_SECRET` must be 64+ chars

```
bash scripts/set-production-env.sh
bash scripts/restart-api.sh
```

Check:

```
bash scripts/verify-secrets.sh
```

5. Redis errors

Symptom: `%22redis://localhost:6379%22` or `Connection is closed`

Cause: `REDIS_URL` has extra quote characters in env file.

Check:

```
grep REDIS_URL packages/backend/.env secrets/providers.env
redis-cli ping
sudo systemctl status redis-server
```

Fix — correct line (no doubled quotes):

```
REDIS_URL=redis://localhost:6379
```

```
sudo systemctl start redis-server
bash scripts/restart-api.sh
```

Good log line: `[Redis] Connected` and `Database and Redis connected – API ready` .

6. Database / migrations

Symptom: `PlanTier` already exists / migration `P3018`

Cause: Baseline migration on existing DB.

```
cd /var/www/sams/packages/backend
npx prisma migrate resolve --rolled-back 20240101000000_init
npx prisma migrate resolve --applied 20240101000000_init
npx prisma migrate deploy
```

Or use:

```
cd /var/www/sams
bash scripts/baseline-prisma-init.sh
cd packages/backend && npx prisma migrate deploy
```

Symptom: Can't connect to database

```
grep DATABASE_URL packages/backend/.env
sudo systemctl status postgresql
```

7. Nginx / uploads / 413 photo error

Symptom: AI says "Could not send that photo" / HTTP 413

Cause: nginx `client_max_body_size` too small on **app** or **super-admin** (not only api subdomain).

Check:

```
grep -n client_max_body_size /etc/nginx/sites-enabled/sams.conf
```

Need **3 lines** at `25m` (app, api, super).

Fix:

```
cd /var/www/sams
git pull origin main
sudo cp nginx/sams.conf /etc/nginx/sites-enabled/sams.conf
sudo nginx -t && sudo systemctl reload nginx
npm run build -w @sams/frontend
npm run build -w @sams/super-admin
```

Symptom: Redirect loop — "page isn't redirecting properly" (Firefox)

Browser shows endless redirects; public URL returns **HTTP/2 301** in a loop. Often appears right after Let's Encrypt / nginx HTTPS setup.

Cause: Cloudflare **Flexible** SSL talks to origin over HTTP, while nginx on port 80 unconditionally redirects to HTTPS (`return 301 https://$host`). Cloudflare and nginx chase each other forever.

Fix (Cloudflare dashboard):

1. Domain → **SSL/TLS** → **Overview**
2. Set encryption mode to **Full (strict)** (not Flexible)
3. Wait ~1 minute; hard-refresh browser or try incognito

Verify on VPS (bypasses Cloudflare; origin must answer 200):

```
curl --resolve app.smart-managment.com:443:127.0.0.1 https://app.smart-managment.com/
```

Expect HTTP **200** and HTML. If this works locally but the public URL still loops → Cloudflare mode is still wrong or cached.

Cleanup: Never leave `*.bak` or `sams-temp.conf` in `/etc/nginx/sites-enabled/` — duplicate server blocks cause odd redirect behavior. Only `sams.conf` should be enabled.

8. AI not working

Symptom: Generic AI errors / not configured

```
bash scripts/verify-secrets.sh --ai-only
bash scripts/diagnose-ai.sh
curl -sS http://127.0.0.1:3001/health | grep -o '"ai":{[^}]*}'
```

Keys live in: `secrets/providers.env` (not git)

Required: `OPENAI_API_KEY` , `OPENAI_BASE_URL` , `OPENAI_MODEL` , optional fallback, `VISION_MODEL` for images.

Groq-only: Image upload needs OpenRouter fallback — Groq text models don't read images.

```
bash scripts/restart-api.sh
```

Symptom: AI talks but won't execute actions (suspend/unsuspend)

- Use explicit commands: `unsuspend school [name]` , `suspend school [name]`
- Destructive actions need reply **yes**
- Super Admin only for platform actions
- School AI actions are role-scoped: HOD/teacher attendance actions only work inside their allowed timetable/session scope, and notification actions are in-app only.
- If AI answers with provider credit/rate-limit errors, code may be healthy but OpenRouter/Groq quota is exhausted. Local DB-backed attendance/timetable answers continue where available.

Symptom: Super Admin AI vs school AI

Panel	URL	Scope
Super Admin AI	super.smart-managment.com	All schools, licenses, suspend
School AI	app.smart-managment.com	One school only

9. SMS / OTP

Current safe mode: app-only notifications

If live SMS capacity or sender approval is not ready, keep school notifications in-app only:

```
cd /var/www/sams
bash scripts/ready-app-only-production.sh
bash scripts/post-deploy-verify.sh
```

Expected: post-deploy verify may warn `SMS sandbox in production` , but it should pass critical checks when `OTP_LOGIN_ENABLED=false` , `OTP_PASSWORD_RESET_ENABLED=false` , and SMS notification sending is disabled in the frontend. Forgot-password SMS self-service is disabled in this mode; use admin reset or configure SMTP/live AT.

Symptom: SMS not delivered

```
curl -sS http://127.0.0.1:3001/health | grep -o '"sms":{[^}]*}'
```

Check `providers.env` :

- `AT_USERNAME=demo1` (dashboard app username, **not** `sams`)
- `AT_API_KEY=atsk_...`
- `AT_SENDER_ID=SAMS` (must be approved in Africa's Talking dashboard)

OTP flags:

- `OTP_LOGIN_ENABLED=false` (login without SMS)
- `OTP_PASSWORD_RESET_ENABLED=true` (forgot-password SMS when sender approved)
- In app-only mode, `ready-app-only-production.sh` sets `OTP_PASSWORD_RESET_ENABLED=false` too.

Until AT approves sender **SAMS**, SMS will fail with `InvalidSenderId` — config can still be correct.

```
bash scripts/configure-production-at.sh
bash scripts/restart-api.sh
```

10. Email

Symptom: `[Email] SMTP not configured / email.configured: false on /health`

Optional for login (keep `OTP_LOGIN_ENABLED=false`). Required for **password-reset email** and OTP-by-email.

Set in `secrets/providers.env` (or `packages/backend/.env`):

- `SMTP_HOST` , `SMTP_PORT` , `SMTP_USER` , `SMTP_PASS`
- `SMTP_FROM_NAME` , `SMTP_FROM_EMAIL`

Gmail: use an [App Password](#), not your normal password.

```
bash scripts/verify-secrets.sh # shows SMTP OK/INFO
bash scripts/restart-api.sh
curl -sS http://127.0.0.1:3001/health | grep -o '"email":{[^}]*}'
```

API returns `503 + EMAIL_NOT_CONFIGURED` when forgot-password needs email but SMTP is missing. In-app notifications still work without email.

Symptom: Old AI threads show decryption errors

If `CONVERSATION_MASTER_KEY` was rotated without keeping the old key:

1. Set `CONVERSATION_MASTER_KEY_PREVIOUS` to the **previous** key (32+ chars) in `providers.env` .
2. `bash scripts/check-conversation-keys.sh`
3. `bash scripts/restart-api.sh`

Users may see a **memory notice** in AI chat for unreadable older messages until threads are re-encrypted.

11. School suspended / users can't login

Symptom: SCHOOL_SUSPENDED / 403

Super Admin — unsuspend via AI:

```
unsuspend school [School Name]
```

Or Super Admin panel → school → Unsuspend.

Check DB:

```
cd /var/www/sams/packages/backend
npx prisma studio
# or SQL: SELECT name, "isSuspended" FROM "School";
```

12. Secrets & env files

File	Purpose	In git?
packages/backend/.env	JWT/refresh/QR/license secrets, DATABASE_URL, PORT	No
secrets/providers.env	AI, AT, SMTP, M-Pesa keys; optional secret overrides	No
secrets/providers.env.example	Template	Yes

Load order: .env → .env.secrets → providers.env (later wins).

Before editing secrets:

```
bash scripts/backup-secrets.sh
# or full backup:
bash scripts/backup-production.sh --with-uploads
```

Repair corrupted providers file:

```
bash scripts/fix-providers-env.sh
bash scripts/verify-secrets.sh
bash scripts/restart-api.sh
```

Rules:

- Never commit real keys
- Never paste shell commands into .env files
- Only # comments and KEY="value" lines
- Production requires distinct 64+ char JWT_SECRET , JWT_REFRESH_SECRET , QR_SECRET , and LICENSE_SECRET values

13. Deploy (full)

Safe updates while live

Use this sequence when schools are actively using SAMS. The deploy script **builds and verifies before touching PM2**, so a failed build leaves the old API running.

```
cd /var/www/sams

# 1. Backup env + database (and uploads if you want)
bash scripts/backup-production.sh --with-uploads

# 2. Optional manual gate (same checks deploy runs before PM2 restart)
bash scripts/pre-deploy-check.sh --skip-tsc

# 3. Full deploy (git reset → npm ci → build → migrate → PM2 → nginx → verify)
bash scripts/deploy-production.sh
```

What deploy does (fail-safe order):

1. Backs up `.env` / secrets (never overwritten by git)
2. `git reset --hard origin/main`
3. `npm ci` + build all packages — **exits here if build fails; PM2 unchanged**
4. `prisma migrate deploy`
5. `pre-deploy-check.sh` (nginx `-t`, SSL, REDIS_URL, dist, startup secrets)
6. PM2 delete + start (only after build OK)
7. Remove nginx `sites-enabled/*.bak`, then `nginx -t` before reload
8. `post-deploy-verify.sh` (health, dist, PM2 online)

Rollback if something looks wrong after deploy:

```
cd /var/www/sams
git log --oneline -5 # find previous commit
git reset --hard <previous-commit-sha> # or: git reset --hard origin/main~1
npm ci && npm run build -w @sams/shared && npm run build -w @sams/backend \
  && npm run build -w @sams/frontend && npm run build -w @sams/super-admin
bash scripts/restart-api.sh
```

Quick API-only restart (no git/build):

```
cd /var/www/sams && bash scripts/restart-api.sh
```

Before deploy:

```
cd /var/www/sams
bash scripts/backup-production.sh
```

Deploy:


```
bash scripts/deploy-production.sh
```

Deploy: backs up env → `git reset --hard origin/main` → rebuilds all packages → prisma migrate → PM2 start → nginx reload → post-deploy verify.

Local dev smoke (before push): with backend running, `bash scripts/smoke-test-local.sh` (optional `VERIFY_LOGIN_*` for `/users/me` + timetable). Full VPS gate: `bash scripts/post-deploy-verify.sh`.

If deploy fails at restart:

```
pm2 start ecosystem.config.js --env production --update-env
pm2 save
```

Never use: `bash scripts/restart-api.sh#`

14. Backup & restore

Create backup:

```
bash scripts/backup-production.sh --with-uploads
ls -la backups/production-*
```

Download to PC (from Windows PowerShell):

```
scp -r root@182.143.228.182:/var/www/sams/backups/production-YYYYMMDD-HHMMSS
"$env:USERPROFILE\Desktop\SAMS-backups\"
```

Restore: See `RESTORE.md` inside backup folder.

15. Super Admin account

```
cd /var/www/sams/packages/backend
npm run create-super-admin
# Reset password:
SUPER_ADMIN_FORCE_RESET=true npm run create-super-admin
```

Login: <https://super.smart-managment.com> — code `SUPERADMIN`.

16. Log locations

Log	Path
PM2 stdout	<code>/var/log/sams/sams-api-out.log</code>
PM2 stderr	<code>/var/log/sams/sams-api-error.log</code>
PM2 live	<code>pm2 logs sams-api</code>

17. Quick command cheat sheet

```
# Status
pm2 status
curl -sS http://127.0.0.1:3001/health

# Restart API (safe)
cd /var/www/sams && bash scripts/restart-api.sh

# Verify secrets
bash scripts/verify-secrets.sh

# Nginx
sudo nginx -t && sudo systemctl reload nginx

# Full deploy
bash scripts/backup-production.sh && bash scripts/deploy-production.sh

# Git update only
cd /var/www/sams && git pull origin main
```

18. Load this runbook into Super Admin AI

Option A — Automatic (after deploy): This file and `docs/SAMS-DEVELOPER-OPS-BOOK-DENIS.md` are injected into Super Admin AI context (ops runbook first, then developer book, then platform docs).

Option B — Knowledge Base (editable in UI):

1. Super Admin → AI Knowledge (or API `POST /api/v1/super/ai-knowledge`)
2. Category: `ops`
3. Title: `SAMS Operations Runbook`
4. Paste sections you use most (keep under ~8000 chars per entry if splitting)

Option C — PDF for your desk: See `docs/README-RUNBOOK-PDF.md`

Deep reference (architecture & failure modes): See `docs/SAMS-DEVELOPER-OPS-BOOK-DENIS.md` and `docs/README-DEVELOPER-OPS-BOOK-PDF.md` — also auto-injected into Super Admin AI after deploy.

19. Symptom index

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Last updated for VPS layout `/var/www/sams` , PM2 `sams-api` , nginx multi-subdomain setup.